

Orientation Discrimination- Signal Detection Task

PSY310: Lab in Psychology

Lab Report

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GitHub Link

<https://github.com/HetanshJalan/Psychology-Experiments>

Introduction

Signal Detection Theory (SDT) is a model for doing this; that is, a theory concerning the distinction between discrimination of a signal and responding to it in a specific manner. In a yes/no detection task, there are four possible results: Hits, Misses, False alarms, Correct rejections. According to SDT, the sensitivity is summarized by d' and the criterion measure summarizes the response bias; the two aspects of performance can be considered separately (Stanislaw & Todorov, 1999).

For this experiment, I used a vertical grating (0°) as the target or signal and any other orientation as noise. Then I made a binary decision with regard to the presence of the target. It was implemented in PsychoPy, an open source program for controlling visual presentation and recording responses (Peirce et al., 2019).

Aim- I wanted to use signal-detection measures to determine how well I could detect the target grating, compared to non-target gratings.

Hypothesis - I thought that the hit rate would be above the false alarm rate and therefore I would get positive sensitivity (d'). This criterion was anticipated to demonstrate if my answers were positively or negatively biased towards reporting the presence or absence of the target.

Method

Participant

The researcher was the participant, 20 year old undergraduate student. The data were taken with the participant label 'final'. This experiment was repeated 100 times.

Materials

The experiment was run in PsychoPy 2026.2.3 on a full screen at 1024 x 768 pixels and with a recorded refresh rate of 60.20 Hz. The computer's processor was an 11th generation Intel Core i3 and the graphics were NVIDIA GeForce MX350. Stimuli consisted of sinusoidal gratings presented on top of a Gaussian mask. The grating was 0.2×0.2 PsychoPy height units, had a contrast of 0.3, and a spatial frequency of 5 cycles per stimulus unit (cpsu).

Each stimulus was preceded by a white fixation cross. The grating was shown for 300ms. I replied with the keyboard with the programmed up and down responses.

Experiment design

A random TrialHandler was used in the experiment. The tilt variable may create either a 0° stimulus or a stimulus between 1° and 5° in either direction. For the signal-detection analysis, I used the condition where the signal was presented 0° as the "signal" condition, and all other non-zero conditions as the "noise" condition. Up responses were coded as the report of the presence of the target (Response = 1) and down responses were coded as the report of the absence of the target (Response = 0).

Parameter	Setting
Software	PsychoPy 2026.2.3
Observations	100 randomized trials
Signal / target	0° vertical grating
Noise	Any non-zero tilt ($\pm 1^\circ$ to $\pm 5^\circ$)
Fixation	White cross, 1,000 ms
Grating duration	300 ms
Grating size	(0.2, 0.2) PsychoPy height units
Spatial frequency	5 cycles per stimulus unit
Contrast	0.3

Mask	Gaussian
Responses	Up = target present; Down = target absent

Table 1- Main signal-detection experiment parameters.

Procedure

I sat at the computer and did 100 random trials. The trials were started with a 1,000ms fixation cross followed by the grating for 300ms. I then programmed in the presence or absence of the target with the keyboard response. The next trial started subsequent to the response.

Data treatment

I categorized each trial from the stimulus and response. A 0° tilt was a target trial and a non 0° tilt was a noise trial. In target trials a target response was scored if the target was hit, and a no-target response was scored if the target was missed. For noise trials, a false alarm was the target response and a no-target response was a correct rejection. I only checked in 1 in each category where I qualified for each trial and left the other 3 category cells empty.

Hit proportion was defined as Hits/target trials and false-alarm proportion was defined as False Alarms / noise trials. Then $d' = z(H) - z(FA)$ and $c = -0.5[z(H) + z(FA)]$ were calculated, according to standard SDT calculations (Stanislaw & Todorov, 1999).

Results

I ran all the 100 trials. Forty-five target trials and 55 noise trials were conducted. I achieved 39 true hits, 6 false misses, 12 false alarms and 43 true rejections.

Outcome	Count	Proportion
Hit	39	0.867
Miss	6	0.133
False alarm	12	0.218
Correct rejection	43	0.782

Table 2- Signal-detection response outcomes.

The hit proportion (H) was 0.867 and the false-alarm proportion (F.A) was 0.218. The resulting sensitivity was $d' = 1.889$. The response criterion was $c = -0.166$. Mean reaction time was 620.5 ms (SD = 113.4), with a median of 611.6 ms.

Table 3. Main signal-detection measures

Measure	Value
Hit proportion	0.867
False-alarm proportion	0.218
d'	1.889
Criterion (c)	-0.166

Table 3- Calculated signal-detection measures.

Discussion

The experimental results gave a sensitivity of $d' = 1.889$. This means there were some separations between the target distribution and the noise distribution was present in my responses, but not very high. My hit proportion was 0.867, while my false alarm proportion was 0.218.

The criterion was $c = -0.166$. The value was negative, meaning that my response pattern was altered so that I might say that the target was present rather than needing more evidence before saying target response. This is also reflected in the raw results: I responded with 39 target-present responses on the signal trials, and 12 target-present responses on the noise trials. Here it is useful to distinguish between this response tendency and sensitivity, rather than lumping all errors together as performance issues (Stanislaw & Todorov, 1999).

The 100-trial data set provides a more useful estimate than the earlier, shorter run, due to the fact that there are many more signal observations than noise observations. This, however, is still one-participant experiment and does not represent an estimate of the population. More samples and multiple runs would be required to explore reliability and individual differences.

Conclusion

Across 100 trials, I obtained a hit proportion of 0.867, a false-alarm proportion of 0.218, $d' = 1.889$, and $c = -0.166$. The results demonstrate that signal detection analysis is able to discriminate between my sensitivity to the target and my tendency to respond that the target was present.

References

- Peirce, J., Gray, J. R., Simpson, S., MacAskill, M., Höchenberger, R., Sogo, H., Kastman, E., & Lindeløv, J. K. (2019). PsychoPy2: Experiments in behavior made easy. Behavior Research Methods, 51(1), 195–203. <https://doi.org/10.3758/s13428-018-01193-y>
- Stanislaw, H., & Todorov, N. (1999). Calculation of signal detection theory measures. Behavior Research Methods, Instruments, & Computers, 31(1), 137–149. <https://doi.org/10.3758/BF03207704>

Appendix-

Trial sheet-

Trial	Tilt	Target	Response	Hit	Miss	FA	CR
1	3	0	0				1
2	-5	0	0				1
3	1	0	1			1	
4	-3	0	0				1
5	2	0	1			1	
6	-2	0	0				1
7	-5	0	0				1
8	0	1	1	1			
9	0	1	1	1			
10	0	1	1	1			
11	0	1	1	1			
12	0	1	1	1			
13	-5	0	0				1
14	0	1	1	1			
15	0	1	1	1			
16	0	1	1	1			
17	0	1	1	1			
18	0	1	1	1			
19	-5	0	0				1

Trial	Tilt	Target	Response	Hit	Miss	FA	CR
20	-3	0	0				1
21	-4	0	0				1
22	-5	0	1			1	
23	0	1	0		1		

24	0	1	1	1			
25	-3	0	0				1
26	0	1	1	1			
27	0	1	1	1			
28	-2	0	1			1	
29	-2	0	1			1	
30	0	1	1	1			
31	0	1	1	1			
32	0	1	1	1			
33	0	1	1	1			
34	0	1	1	1			
35	2	0	1			1	
36	0	1	1	1			
37	4	0	1			1	
38	0	1	1	1			
39	4	0	0				1
40	0	1	1	1			
41	-2	0	1			1	
42	0	1	1	1			
43	0	1	1	1			
44	4	0	0				1
45	3	0	0				1
46	0	1	1	1			
47	3	0	0				1
48	0	1	1	1			
49	3	0	0				1
50	-3	0	0				1
51	0	1	0		1		
52	0	1	1	1			
53	-4	0	1			1	
54	-2	0	0				1
55	1	0	1			1	
56	-2	0	0				1
57	0	1	0		1		
58	-5	0	0				1
59	3	0	0				1

60	0	1	0		1		
61	3	0	0				1

Trial	Tilt	Target	Response	Hit	Miss	FA	CR
62	-5	0	0				1
63	2	0	0				1
64	2	0	0				1
65	-4	0	0				1
66	-5	0	0				1
67	0	1	0		1		
68	-3	0	0				1
69	0	1	1	1			
70	1	0	1			1	
71	0	1	1	1			
72	0	1	1	1			
73	4	0	0				1
74	4	0	0				1
75	0	1	1	1			
76	2	0	0				1
77	-5	0	0				1
78	-2	0	1			1	
79	-5	0	0				1
80	0	1	1	1			
81	0	1	1	1			
82	0	1	1	1			
83	0	1	0		1		
84	4	0	0				1
85	-5	0	0				1
86	0	1	1	1			
87	-3	0	0				1
88	0	1	1	1			
89	4	0	0				1
90	-4	0	0				1
91	0	1	1	1			
92	0	1	1	1			

93	-4	0	0				1
94	4	0	0				1
95	0	1	1	1			
96	0	1	1	1			
97	-5	0	0				1
98	-4	0	0				1
99	-2	0	0				1
100	-5	0	0				1

Table - Trial-level target/response coding and signal-detection outcome. Blank cells mean that the outcome did not apply.